RETAIL SALES EXCERCISE

① SELECT STATEMENT

QUERY: DISPLAY ALL COLUMNS FOR ALL TRANSACTIONS LIMIT 10

```
# SELECT *
FROM `WORKSPACE`.`DEFAULT`.`RETAIL_SALES_DATASET`
LIMIT 10;
```

TRANSACTION ID	DATE	CUSTOMER ID	GENDER	AGE	PRODUCT CATEGORY	QUANTITY	TOTAL AMOUNT
1	2023-11-24	CUST001	MALE	34	BEAUTY	3	150
2	2023-02-27	CUST002	FEMALE	26	CLOTHING	2	1000
3	2023-01-13	CUST003	MALE	50	ELECTRONICS	1	30
4	2023-05-21	CUST004	MALE	37	CLOTHING	1	500
5	2023-05-06	CUST005	MALE	30	BEAUTY	2	100
6	2023-04-25	CUST006	FEMALE	45	BEAUTY	1	30
7	2023-03-13	CUST007	MALE	46	CLOTHING	2	50
8	2023-02-22	CUST008	MALE	30	ELECTRONICS	4	100
9	2023-12-13	CUST009	MALE	63	ELECTRONICS	2	600
10	2023-10-07	CUST010	FEMALE	52	CLOTHING	4	200

② DISPLAY ONLY THE TRANSACTION ID, DATE, CUSTOMER ID FOR ALL RECORDS.

QUERY: TRANSACTION ID, DATE & CUSTOMER ID

```
# SELECT `TRANSACTION ID`, `DATE`, `CUSTOMER ID`
FROM `WORKSPACE`.`DEFAULT`.`RETAIL_SALES_DATASET`;
```

TRANSACTION ID	DATE	CUSTOMER ID
1	2023-11-24	CUST001
2	2023-02-27	CUST002
3	2023-01-13	CUST003
4	2023-05-21	CUST004
5	2023-05-06	CUST005
6	2023-04-25	CUST006

ECT. ↓

NB I LIMITED TO 6

③ SELECT DISTINCT STATEMENT

QUERY: DISPLAY ALL THE DISTINCT PRODUCT CATEGORIES IN THE DATASET

```
# SELECT DISTINCT 'PRODUCT CATEGORY'
FROM 'WORKSPACE', 'DEFAULT', 'RETAIL_SALES_DATASET';
```

PRODUCT CATEGORY
BEAUTY
CLOTHING
ELECTRONICS

④ DISPLAY ALL THE DISTINCT GENDER VALUES IN THE DATASET

QUERY: GENDER

```
# SELECT DISTINCT 'GENDER'
FROM 'WORKSPACE', 'DEFAULT', 'RETAIL_SALES_DATASET';
```

GENDER
MALE
FEMALE

⑥ WHERE CLAUSE

QUERY: DISPLAY ALL TRANSACTION WHERE THE PRICE PER UNIT IS BETWEEN 100 AND 500

```
# SELECT 'PRICE PER UNIT' OR '*'
FROM 'WORKSPACE'.'DEFAULT'.'RETAIL_SALES_DATASET'
WHERE PRICE PER UNIT;
```

TRANSACTION ID	DATE	CUSTOMER ID	GENDER	AGE	PRODUCT CATEGORY	QUANTITY	PRICE PER UNIT
2	2023-02-27	CUST002	FEMALE	26	CLOTHING	2	500
4	2023-05-21	CUST004	MALE	37	CLOTHING	1	500
9	2023-12-13	CUST009	MALE	63	ELECTRONICS	2	300
13	2023-08-05	CUST013	MALE	22	ELECTRONICS	3	500
15	2023-01-16	CUST015	FEMALE	42	ELECTRONICS	4	500

ECT. ↓ NB I LIMITED TO 5

⑤ DISPLAY ALL TRANSACTION AGE IS GREATER THAN 40.

QUERY: ALL COLUMNS

```
# SELECT *
FROM 'WORKSPACE'.'DEFAULT'.'RETAIL_SALES_DATASET'
WHERE AGE > 40;
```

TRANSACTION ID	DATE	CUSTOMER-ID	GENDER	AGE	PRODUCT CATEGORY	QTY	TOTAL AMOUNT
1	2023-01-13	CUST003	MALE	50	ELECTRONICS	1	30
2	2023-04-25	CUST006	FEMALE	45	BEAUTY	1	30
3	2023-03-13	CUST007	MALE	46	CLOTHING	2	50
4	2023-12-13	CUST009	MALE	63	ELECTRONICS	2	600
5	2023-10-07	CUST010	FEMALE	52	CLOTHING	4	200
6	2023-10-17	CUST014	MALE	64	CLOTHING	4	120
7	2023-01-16	CUST015	FEMALE	42	ELECTRONICS	4	2000

ECT ↓ NB I LIMITED TO 7

⑥ DISPLAY ALL TRANSACTIONS WHERE THE PRODUCT CATEGORY IS EITHER BEAUTY OR ELECTRONICS

QUERY: ALL COLUMNS

```
# SELECT *
FROM 'WORKSPACE'.'DEFAULT'.'RETAIL_SALES_DATASET'
WHERE 'PRODUCT CATEGORY' IN ('BEAUTY', 'ELECTRONICS');
```

TRANSACTION ID	DATE	CUSTOMER ID	GENDER	AGE	PRODUCT CATEGORY	QTY	PRICE PER UNIT	TOTAL AMOUNT
1	2023-11-24	CUST001	MALE	34	BEAUTY	3	50	150
5	2023-05-06	CUST005	MALE	30	BEAUTY	2	50	100
6	2023-04-25	CUST006	FEMALE	45	BEAUTY	1	30	30
3	2023-01-13	CUST003	MALE	50	ELECTRONIC	1	30	30
8	2023-02-22	CUST008	MALE	30	ELECTRONIC	4	25	100
9	2023-12-13	CUST009	MALE	63	ELECTRONIC	2	300	600
12	2023-10-30	CUST012	MALE	35	BEAUTY	3	25	75
13	2023-08-05	CUST013	MALE	22	ELECTRONIC	3	500	1500

ECT ↓ NB I LIMITED TO 8

(8) DISPLAY ALL TRANSACTIONS WHERE THE PRODUCT CATEGORY IS NOT 'CLOTHING'
QUERY : ALL COLUMNS

```
# SELECT *
FROM `WORKSPACE`.`DEFAULT`.`RETAIL_SALES_DATASET`
WHERE `PRODUCT CATEGORY` != 'CLOTHING';
```

TRANSACTION ID	DATE	CUSTOMER ID	GENDER	AGE	PRODUCT CATEGORY	QTY	PRICE PER UNIT	TOTAL AMOUNT
1	2023-11-24	CUST001	MALE	34	BEAUTY	3	50	150
3	2023-01-13	CUST003	MALE	50	ELECTRONICS	1	30	30
5	2023-05-06	CUST005	MALE	30	BEAUTY	2	50	100
6	2023-04-25	CUST006	FEMALE	45	BEAUTY	1	30	30
8	2023-02-22	CUST008	MALE	30	ELECTRONICS	4	25	100
9	2023-12-13	CUST009	MALE	63	ELECTRONICS	2	300	600
12	2023-10-30	CUST012	MALE	35	BEAUTY	3	25	75
13	2023-08-05	CUST013	MALE	22	ELECTRONICS	3	500	1500

ECT ↓

NB I LIMITED TO 8

(9) DISPLAY ALL TRANSACTION WHERE THE QUANTITY IS GREATER THAN OR EQUAL TO 3.
QUERY : ALL COLUMNS

```
# SELECT *
FROM `WORKSPACE`.`DEFAULT`.`RETAIL_SALES_DATASET`
WHERE `QUANTITY` > 3;
```

TRANSACTION ID	DATE	CUSTOMER ID	GENDER	AGE	PRODUCT CATEGORY	QTY	PRICE PER UNIT	TOTAL AMOUNT
8	2023-02-22	CUST008	MALE	30	ELECTRONICS	4	25	100
10	2023-10-07	CUST010	FEMALE	52	CLOTHING	4	50	200
14	2023-01-17	CUST014	MALE	64	CLOTHING	4	30	120
15	2023-01-16	CUST015	FEMALE	42	ELECTRONICS	4	500	2000
17	2023-04-22	CUST017	FEMALE	27	CLOTHING	4	25	100
23	2023-04-12	CUST023	FEMALE	35	CLOTHING	4	30	120
31	2023-05-23	CUST031	MALE	44	ELECTRONICS	4	300	1200
38	2023-03-21	CUST038	MALE	38	BEAUTY	4	50	200

ECT ↓

(10) AGGREGATE FUNCTION

QUERY: COUNT THE TOTAL NUMBER OF TRANSACTIONS

```
# SELECT COUNT (*)
FROM 'WORKSPACE'.'DEFAULT'.'RETAIL_SALES_DATASET';
```

TOTAL_TRANSACTIONS
1000

(11) FIND THE AVERAGE OF CUSTOMERS

QUERY: AVERAGE_AGE

```
# SELECT AVG (AGE)
FROM 'WORKSPACE'.'DEFAULT'.'RETAIL_SALES_DATASET';
```

AVERAGE_AGE
41,39

(12) FIND THE TOTAL QUANTITY OF PRODUCTS SOLD

QUERY: TOTAL_QUANTITY

```
# SELECT SUM (QUANTITY) AS TOTAL_QUANTITY
— FROM 'WORKSPACE'.'DEFAULT'.'RETAIL_SALES_DATASET';
```

TOTAL_QUANTITY
2514

(13) FIND THE MAXIMUM TOTAL AMOUNT SPENT IN SINGLE TRANSACTION

QUERY: MAX_TOTAL_AMOUNT

```
# SELECT MAX ('TOTAL AMOUNT') AS MAX_TOTAL_AMOUNT
FROM 'WORKSPACE','DEFAULT','RETAIL_SALES_DATASET';
```

MAX_TOTAL_AMOUNT
2000

(14) FIND THE MINIMUM PRICE PER UNIT IN THE DATASET
QUERY: MIN_PRICE_PER_UNIT

```
# SELECT MIN('PRICE PER UNIT') AS MIN_PRICE_PER_UNIT
FROM 'WORKSPACE', 'DEFAULT'.'RETAIL_SALES_DATASET';
```

MIN_PRICE_PER_UNIT
25

(15) GROUP BY STATEMENT
FIND THE NUMBER OF TRANSACTIONS PER PRODUCT CATEGORY

```
# SELECT PRODUCT CATEGORY, COUNT('TRANSACTION ID')
FROM 'WORKSPACE', 'DEFAULT', 'RETAIL_SALES_DATASET';
```

COUNT('TRANSACTION ID')	PRODUCT CATEGORY
307	BEAUTY
351	CLOTHING
342	ELECTRONIC

(16) FIND THE TOTAL REVENUE ('TOTAL AMOUNT') PER GENDER
QUERY: GENDER, TOTAL_REVENUE

```
# SELECT 'GENDER', SUM('TOTAL AMOUNT')
FROM 'WORKSPACE', 'DEFAULT', 'RETAIL_SALES_DATASET'
GROUP BY 'GENDER';
```

SUM('TOTAL AMOUNT')	GENDER
223160	MALE
232840	FEMALE

(17) FIND THE AVERAGE PRICE PER UNIT PER PRODUCT CATEGORY

QUERY: PRODUCT CATEGORY, AVERAGE_PRICE

```
# SELECT AVG('PRICE PER UNIT') AS AVERAGE_PRICE, PRODUCT CATEGORY
FROM 'WORKSPACE', 'DEFAULT', 'RETAIL_SALES_DATASET'
GROUP BY PRODUCT CATEGORY;
```

AVERAGE_PRICE	PRODUCT CATEGORY
184.06	BEAUTY
174,29	CLOTHING
181,9	ELECTRONICS

(18) HAVING CLAUSE

FIND THE TOTAL REVENUE PER PRODUCT CATEGORY WHERE TOTAL REVENUE IS GREATER THAN 10 000.

```
# SELECT SUM('TOTAL AMOUNT') AS TOTAL_REVENUE, 'PRODUCT CATEGORY'
FROM 'WORKSPACE', 'DEFAULT', 'RETAIL_SALES_DATASET'
GROUP BY 'PRODUCT CATEGORY'
HAVING SUM('TOTAL AMOUNT') > 10000;
```

TOTAL_REVENUE	PRODUCT CATEGORY
143515	BEAUTY
155580	CLOTHING
156905	ELECTRONIC

(19) FIND THE AVERAGE QUANTITY PER PRODUCT CATEGORY WHERE THE AVERAGE IS MORE THAN 2. QUERY: PRODUCT_CATEGORY, AVERAGE_QUANTITY

```
# SELECT AVG('QUANTITY') AS AVERAGE_QUANTITY, 'PRODUCT CATEGORY'
FROM 'WORKSPACE','DEFAULT','RETAIL_SALES_DATASET'
GROUP BY 'PRODUCT CATEGORY'
HAVING AVG('QUANTITY') > 2;
```

AVERAGE_QUANTITY	PRODUCT CATEGORY
2,51	BEAUTY
2,55	CLOTHING
2,48	ELECTRONICS

(20) CASE STATEMENT

DISPLAY A COLUMN CALLED SPENDING_LEVEL THAT SHOWS 'HIGH' IF TOTAL AMOUNT > 1000, OTHERWISE 'LOW'.

TRANSACTION ID, TOTAL AMOUNT, SPENDING_LEVEL

```
# SELECT 'TRANSACTION ID',
        'TOTAL AMOUNT',
      CASE
        WHEN 'TOTAL_AMOUNT' > 1000 THEN 'HIGH'
        ELSE 'LOW'
      END AS SPENDING_LEVEL
FROM 'WORKSPACE'.'DEFAULT'.'RETAIL_SALES_DATASET';
```

TRANSACTION ID	TOTAL AMOUNT	SPENDING_LEVEL
1	150	LOW
2	1000	LOW
3	30	LOW
4	500	LOW
5	100	LOW
6	30	LOW

NB: I LIMITED TO 6

ECT ↓

(21) DISPLAY A NEW COLUMN CALLED AGE_GROUP THAT LABELS CUSTOMER AS

- 'YOUTH' IF AGE < 30
- 'ADULT' IF AGE IS BETWEEN 30 AND 59
- 'SENIOR' IF AGE >= 60

```
# SELECT 'CUSTOMER ID',
        AGE,
    CASE
        WHEN AGE < 30 THEN 'YOUTH'
        WHEN AGE BETWEEN 30 AND 59 THEN 'ADULT';
```

CUSTOMER ID	AGE	AGE_GROUP
CUST001	34	ADULT
CUST002	26	YOUTH
CUST003	50	ADULT
CUST004	37	ADULT
CUST005	30	ADULT
CUST006	45	ADULT
CUST007	46	ADULT
CUST008	30	ADULT

ECT ↓

NB : I LIMITED TO 8